

$$* P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$* P(A \cap B) = P(A|B) \cdot P(B) \text{ [Product Rule]}$$

$$* P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \rightarrow \text{Bayes Theorem}$$

* Guilty or not?

$$\rightarrow P(G|C) = 0.98, \quad P(G'|C) = 0.02$$

$$P(G'|C') = 0.97, \quad P(G|C') = 0.03$$

$$P(C) = 0.008, \quad P(C') = 0.992$$

Find, $P(C|G)$ and $P(C'|G)$, compare.

$$P(C|G) = \frac{P(G|C) \cdot P(C)}{P(G)}$$

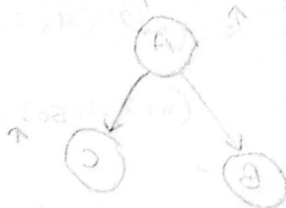
$$P(C'|G) = \frac{P(G|C') \cdot P(C')}{P(G)}$$

we don't need to find $P(G)$,
we can just compare numerator
whichever is bigger.

Now calculation is substantially reduced.

Chain Rule:

$$P(A, B, C, D, E) \rightarrow P(A|B \cap C \cap D \cap E) \cdot P(B \cap C \cap D \cap E)$$



$$\begin{cases} \uparrow 5.0 = (5/10) \\ \downarrow 5.0 = (5/10) \end{cases}$$

$$\begin{aligned} P(A) &= P(A) \\ P(B|A) &= P(B|A) \\ P(B) &= P(B) \\ P(B|A) \cdot P(A) &= P(B \cap A) \\ P(B) \cdot P(A) &= P(B \cap A) \end{aligned}$$

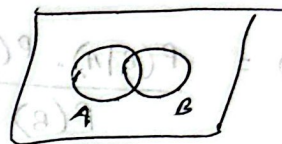
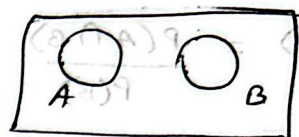
* Independance

A, B

Dependency,

- Probability distribution - A → changes

↳ Prob dist. - B → changes



- Knowing A → additional knowledge → B | B → A

- May or may not have a cause and effect relationship.

R	0.2
R'	0.8

if friend calls and tell its raining,

U	0.3
U'	0.7

$$P(R) = 0.2$$

$$P(R') = 0.8$$

$$P(U) = 0.3$$

$$P(U') = 0.7$$

$$\rightarrow P(U|R) = 0.9$$

When it changes, U also changes.

R and U are dependant.

Prior,

$$P(\text{blue}) = 0.2$$

$$P(\text{blue}') = 0.8$$

doesn't change U

$$\rightarrow P(U|B) = 0.3$$

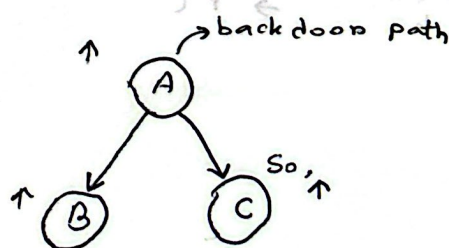
if independant,

$$P(A|B) = P(A)$$

$$P(B|A) = P(B) \rightarrow \text{if indep.}$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

$$\therefore P(A \cap B) = P(A) \cdot P(B)$$



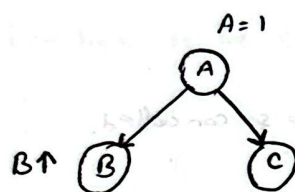
However, B and C are dependant.

A, B
if independent,
→ given 3rd event C

$$P(A \cap B | C) = P(A | C) \cdot P(B | C)$$

$$P(A | B \cap C) = P(A | C)$$

$$P(B | A \cap C) = P(B | C)$$



$C?$ it won't change since A is fixed.

B and C are now conditionally independent.

Naive Bayes Classifier:

$$y = f(x)$$

$$y = f(x_1, x_2, x_3, x_4)$$

↳ All given, except the function.

It thinks that the input features are conditionally independent, given the output.

Find the conditional probabilities.

$$\text{Let, } 1) P(PT = y | S, C, H, Str) = \frac{P(S, C, H, Str | y) \cdot P(y)}{P(S, C, H, Str)}$$

$$2) P(PT = n | S, C, H, Str) = \frac{P(S, C, H, Str | n) \cdot P(n)}{P(S, C, H, Str)}$$

> Denom = same
check num and compare.

Naive Bayes assuming: S, C, H, Str are conditionally independent given PT .

$$1) = P(S | y) \cdot P(C | y) \cdot P(H | y) \cdot P(Str | y) \cdot P(y)$$

$$2) = P(S | n) \cdot P(C | n) \cdot P(H | n) \cdot P(Str | n) \cdot P(n)$$

* $P(HIV) = 0.008$

$P(T|HIV) = 95\% = 0.95$, $P(T'|HIV) = 0.05$

$P(T'|HIV') = 95\% = 0.95$, $P(T|HIV') = 0.05$

Compare ,

$P(HIV|T_1 \cap T_2)$

$P(HIV'|T_1 \cap T_2)$

What's more likelier.

$\Rightarrow P(T_1 \cap T_2 | HIV) \cdot P(HIV)$

Denom will be same so cancelled.

$\Rightarrow P(T_1 \cap T_2 | HIV') \cdot P(HIV')$

$= P(T_1 | HIV') \cdot P(T_2 | HIV') \cdot P(HIV')$

T_1 and T_2 are independent of each other.

$P(y|A, B, C, D) = \frac{P(A, B, C, D|y) \cdot P(y)}{P(A, B, C, D)}$

$= \frac{P(A|y) \cdot P(B|y) \cdot P(C|y) \cdot P(D|y) \cdot P(y)}{P(A) \cdot P(B) \cdot P(C) \cdot P(D)}$

$P(T_1 \cap T_2) =$

$(5) \cdot (5) \cdot (5) \cdot (5) \cdot (5)$

$(12, 11, 12, 12) \cdot (2) \cdot (2)$

$(11) \cdot (11) \cdot (11) \cdot (11) \cdot (11)$

$(12, 11, 12, 12) \cdot (2) \cdot (2)$

$(5) \cdot (5) \cdot (5) \cdot (5) \cdot (5)$

$(11) \cdot (11) \cdot (11) \cdot (11) \cdot (11)$

Linear Regression & Gradient Descent:

X	Y
1	4
2	5
3	6
?	?

Assume Y is a quantitative variable.

" X and Y has some linear relationship.

To find the best fit straight line.

Loss Functions.

Gradient descent can get stuck in local minima.

Function has to be continuous.

SSR

$$\sum_{i=0}^n (y_i - \hat{y}_i)^2$$

Actual Predicted

Gradient Descent is a local search technique.

∴ SSR,

$$x=1, y_1=4$$

$$y'_1 = 0.1(1) + 0.6$$

$$x=2, y_2=5$$

$$y'_2 = 0.1(2) + 0.6$$

$$= 0.8$$

$$x=3, y_3=6$$

$$y'_3 = 0.1(3) + 0.6$$

$$= 0.9$$

$$\{y_1 - (m_1 x_1 + c)\}^2 + \{y_2 - (m_1 x_2 + c)\}^2 + \{y_3 - (m_1 x_3 + c)\}^2$$

for all params.

$$m_{\text{new}} = m_{\text{old}} - \text{Step Size}$$

Learning Rate

$$\text{Step Size} = \frac{dSSR}{dm_{\text{old}}} \times L.R$$

Need to do this for c as well.

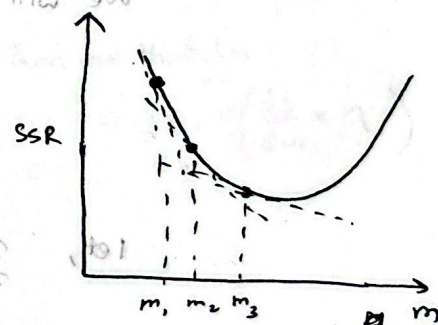
$$\frac{dSSR}{dm_1} = 2\{y_1 - (m_1 x_1 + c)\} \{-x_1\} - 2\{y_2 - (m_1 x_2 + c)\} \{-x_2\} - 2\{y_3 - (m_1 x_3 + c)\} \{-x_3\}$$

$$= -2.1$$

$$m_{\text{new}} = 0.1 - \{-2.1 \times 0.1\} = 0.31$$

2nd iter, $m_2 = 0.3$
 $c_2 = 0.7$ } Now check loss again.

$$c_{\text{new}} = 0.6 - \{-1.5 \times 0.1\} = 0.75$$



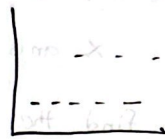
Using Gradient Descent.

$$\frac{dSSR}{dm_1}$$

$$\frac{dSSR}{dc_1} = 2\{y_1 - (m_1 x_1 + c)\} \{-1\} + 2\{y_2 - (m_1 x_2 + c)\} \{-1\} + 2\{y_3 - (m_1 x_3 + c)\} \{-1\}$$

Logistic Regression & Gradient Descent:

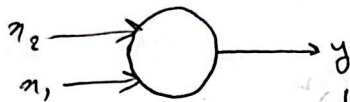
x_2	x_1	y
3	1	Yes - 1
4	2	No - 0
5	3	Yes - 1



Used for binary classification problems mainly.

MLP

Perceptron is a function



* Gradient descent can only optimize continuous functions.

These are likelihoods.

① Regression

$w_1x_1 + w_2x_2 + \beta$ } it is a linear equation.

output will also be linear.

↳ goes from $-\infty$ to ∞

↳ So this needs to pass into a non-linear functions.

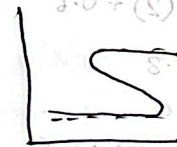
[activation function]

② ** A sigmoid func is any function that creates a S shaped curve

For Logistic Regression,

We need to find the best-fitting S curve.

We will be using a logistic function



activation func.

$$a = \frac{1}{1 + e^{-z}} \quad z = w_1x_1 + w_2x_2 + \beta$$

↳ value is bounded between (0, 1)

let, $x_2 = 5$, $x_1 = 7$ output predicts $\hat{y} = 0.7$

"Having a threshold ≥ 0.5 "

Here Loss function is BCE [Binary Cross Entropy]

$$- [y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$$

when $y=1$, $- [y \log(\hat{y})]$

" $y=0$, $- [(1-y) \log(1-\hat{y})]$

others become zero.

Learning Rate

$= \eta$

It helps penalizes cases where we are Confidently wrong.

After calculating losses, change params.

$$w_{new} = w_{old} - \text{step size } r$$

$$\left(LR \times \frac{d BCE}{dw_1} \right)$$

same for other params.

$\frac{dL}{dw_1} \rightarrow$ Chain Rule
 $\rightarrow$ used in composite functions.

For $\frac{dL}{dw_1}$,

$$a = f(g(n))$$

$$\frac{d f(g(n))}{dx} = f'(g(n)) \cdot g'(n)$$

$$L \rightarrow a$$

$$L = f_1(a)$$

$$a = f_2(z)$$

$$z = f_3(w_1)$$

$$f_1(f_2(f_3(w_1)))$$

$$\frac{dL}{dw_1} = \frac{dL}{da} \times \frac{da}{dz} \times \frac{dz}{dw_1}$$

$$\frac{dL}{dw_2} = \frac{dL}{da} \times \frac{da}{dz} \times \frac{dz}{dw_2}$$

$$\frac{dL}{d\beta} = \frac{dL}{da} \times \frac{da}{dz} \times \frac{dz}{d\beta}$$

x_1	x_2	L
1	1	1
1	1	1
1	1	1
1	1	1

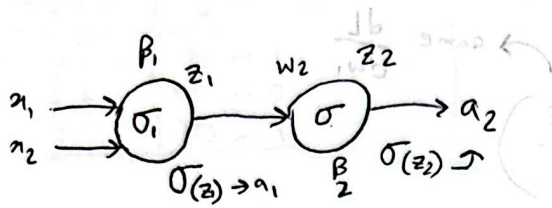
all $\frac{dL}{dw_1}$

Then find average $\frac{dL}{dw_1}$

Then use that in

$$w_1 = w_1 - \left\{ \frac{dL}{da} \times \frac{da}{dz} \times \frac{dz}{dw_1} \times \eta \right\}$$

Now a larger network.



$$L = f_1(a_2)$$

$$a = f(z)$$

$$z = f(w)$$

$$L = f_1(f_2(f_3(w)))$$

outermost

L func.

Backpropagation,

$$w_2 = w_2 - \left\{ \frac{dL}{da_2} \times \frac{da_2}{dz_2} \times \frac{dz_2}{dw_2} \times \eta \right\}$$

$$w_1 = w_1 - \left\{ \frac{dL}{dw_1} \times \eta \right\}$$

z_2 is a func of w_2 . But w_2 is not of anything. Can't use $\frac{dz_2}{dw_2}$ to complete the chain. We know, z_2 is also

$$\left\{ \frac{dL}{da_2} \times \frac{da_2}{dz_2} \times \frac{dz_2}{da_1} \times \frac{da_1}{dz_1} \times \frac{dz_1}{dw_1} \right\}$$

a function of a_1 . So we use

We do start from the back/last [backpropagation]

because last layer already calculates $\frac{dL}{da_2}$, $\frac{da_2}{dz_2}$. We store them

and we propagate them backwards.

Better optimization.

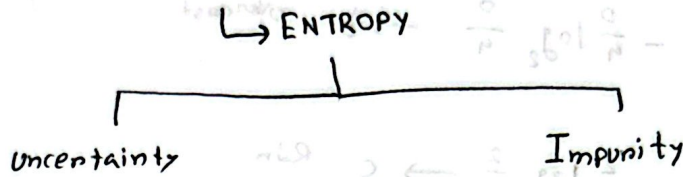
Cycle of Forward Propagation to calculate the predicted value.

$$\left\{ \eta \times \frac{5b}{1wb} \times \frac{1b}{5b} \times \frac{1b}{1wb} \right\} = 1wb$$

Decision Trees :

{ID3 only}

We need to find information gain (IG)



① 3H
3T

Highest uncertainty

② 6H
0T

highest certainty

Least uncertainty

③ 4H
2T

$E = 0$

Entropy calculation,

e.g.

3H
3T

$$-\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) = 1$$

this is highest entropy.

lowest certainty,
highest uncertainty

Prior Entropy

Conditional Entropy

Difference in the entropies.

$$n - y = IG$$

Need to find least uncertainty.

Reduction in entropy.

Sample dataset,

ID 3,

[Reducing uncertainty in 'Decision']

$$IG(\text{Outlook}) \rightarrow E(D) - E(D|\text{outlook})$$

Prior entropy $\rightarrow$ conditional entropy

$$E(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} \rightarrow X$$

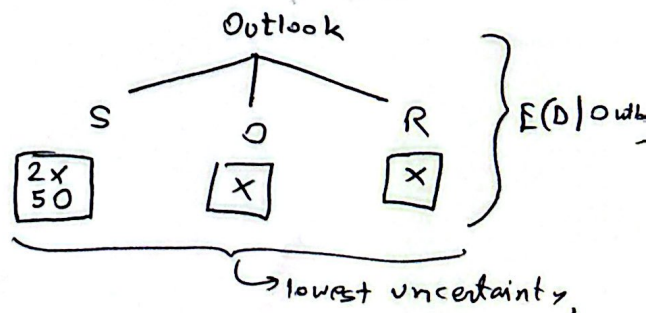
$$E(D|\text{Outlook} = \text{Sunny})$$

$$\left. \begin{array}{l} + \\ E(D|\text{Outlook} = \text{overcast}) \\ + \\ E(D|\text{Outlook} = \text{Rain}) \end{array} \right\} \text{ gives } E(D|\text{Outlook})$$

$$\text{e.g. } -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5}$$

e.g.,

High uncertainty
[no info of outlook]



$$\Rightarrow -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \rightarrow a \quad \text{sum}$$

$$\Rightarrow -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \rightarrow b \quad \text{overcast}$$

$$\Rightarrow -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \rightarrow c \quad \text{rain}$$

Total 14 Samples.

$$Y = a + b + c$$

gives equal priority to all [Wrong]
probabilities.

$$Y = w_1 a + w_2 b + w_3 c$$

$$Y = \frac{5}{14} a + \frac{4}{14} b + \frac{5}{14} c \quad \therefore X - Y$$

$$1 = \left(\frac{1}{2}\right) \log_2 \frac{1}{2} - \left(\frac{1}{2}\right) \log_2 \frac{1}{2} -$$

$$IG(\text{Temp}) \rightarrow E(D) - E(D|\text{Temp})$$

$$E(D) = X \quad \{\text{previously calculated}\}$$

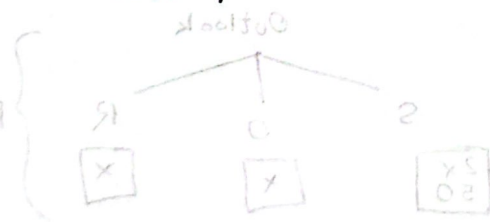
$$E(D|\text{Hot}) = -\frac{2}{7} \dots - \frac{2}{7} \dots \rightarrow a$$

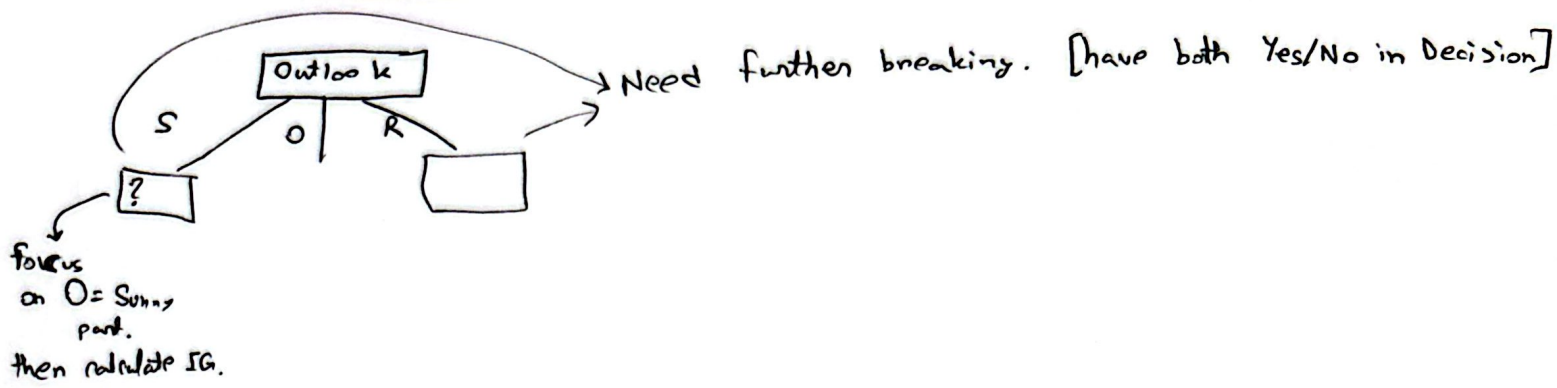
$$E(D|\text{Mild}) = -\frac{4}{6} \dots - \frac{2}{6} \dots \rightarrow b$$

$$E(D|\text{Cool}) = -\frac{3}{4} \dots - \frac{1}{4} \dots \rightarrow c$$

$$Y = \frac{4}{14} a + \frac{6}{14} b + \frac{4}{14} c$$

$$\therefore X - Y$$





IG (Temp) $\rightarrow$ O = Sunny

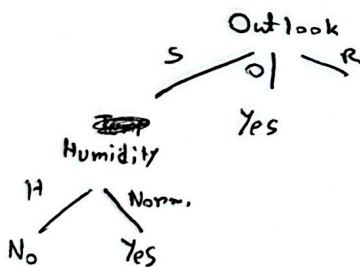
$$E(D) \rightarrow -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \rightarrow x$$

$$E(D|Hot) = 0$$

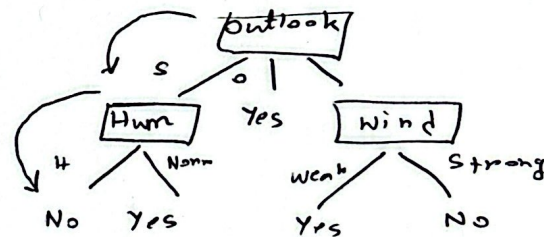
$$E(D|mild) = 1 \quad \therefore y = \frac{2}{5}(0) + \frac{1}{5}(0) + \frac{2}{5}(1) = \frac{2}{5}$$

$$E(D|cool) = 0 \quad \therefore x - y$$

Humidity has $E \rightarrow 0$.



$\rightarrow$



$\therefore$ O = S
T = C
H = H
W = Str.

$\rightarrow$ Trace tree,